

# Complete Git Setup Guide for Windows CMD

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## 1. What is Git?

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**Git** is a distributed version control system that tracks changes in your files (usually code). It allows multiple people to collaborate efficiently on the same project.

## 2. Installing Git

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1. Visit <https://git-scm.com/>
2. Download the latest version for Windows
3. Run the installer
4. During setup, **check the box** to “Add Git Bash to PATH”
5. Finish installation

## 3. Check Git Installation

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Open **Command Prompt (CMD)** or **PowerShell** and type:

```
git --version
```

Output example:

```
git version 2.45.0.windows.1
```

If you see this, Git is correctly installed.

## 4. Setting Up Git for the First Time

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Git needs your identity (name and email) for commits.

```
git config --global user.name "Your Name"
git config --global user.email "youremail@example.com"
```

- `--global` means this applies to all repositories on your computer.
- Omit `--global` if you want to set it only for a specific repository.

### Example

```
git config --global user.name "Jane Doe"
git config --global user.email "jane@example.com"
```

## 5. Verify Your Git Configuration

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To see all settings:

```
git config --list
```

To check specific values:

```
git config user.name
git config user.email
```

## 6. Set Default Branch Name

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Modern Git prefers **main** instead of **master**.

Set it once globally:

```
git config --global init.defaultBranch main
```

## 7. Manage Git Credentials

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If you use HTTPS for GitHub, GitLab, etc., you can store login info.

### Option 1: Store credentials in plain text

```
git config --global credential.helper store
```

### Option 2: Use Windows Credential Manager (recommended)

```
git config --global credential.helper manager
```

## 8. Configure Proxy (if needed)

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If you are behind a corporate proxy, use:

```
git config --global http.proxy http://proxyuser:proxypassword@proxy.s
```

Unset proxy:

```
git config --global --unset http.proxy
```

## 9. Set Git Path in Environment Variables (if Git not recognized)

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If `git` command doesn't work in CMD:

1. Press **Win + S** → type **Environment Variables**
2. Open **Edit the system environment variables**
3. Click **Environment Variables**
4. Under **System Variables**, find `Path` → click **Edit**
5. Add this path (default Git install path):

```
C:\Program Files\Git\bin
C:\Program Files\Git\cmd
```

6. Click **OK** → Restart CMD

Now try:

```
git --version
```

## 10. Initialize a New Git Repository

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Create a new project:

```
mkdir myproject
cd myproject
git init
```

Creates a hidden `.git` folder — your repository.

## 11. Basic Git Commands

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| Command                              | Description              |
|--------------------------------------|--------------------------|
| <code>git init</code>                | Initialize a repository  |
| <code>git clone &lt;url&gt;</code>   | Clone a remote repo      |
| <code>git status</code>              | Check file changes       |
| <code>git add &lt;file&gt;</code>    | Stage a file for commit  |
| <code>git add .</code>               | Stage all changes        |
| <code>git commit -m "message"</code> | Commit with a message    |
| <code>git log</code>                 | View commit history      |
| <code>git push origin main</code>    | Upload commits to remote |

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| Command                           | Description             |
|-----------------------------------|-------------------------|
| <code>git pull origin main</code> | Download latest changes |

## 12. View, Edit, or Remove Configuration

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View all configs:

```
git config --list
```

Edit config manually:

```
git config --global --edit
```

Unset a config:

```
git config --global --unset user.name
```

## 13. Useful Global Aliases (optional)

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Make Git commands shorter:

```
git config --global alias.st status
git config --global alias.cm "commit -m"
git config --global alias.co checkout
git config --global alias.br branch
git config --global alias.lg "log --oneline --graph --all"
```

Now you can type:

```
git st
git cm "message"
git lg
```

## 14. Backup and Export Git Config

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To export your global config file:

```
notepad "%USERPROFILE%\.gitconfig"
```

You can copy this file to another system to keep your Git setup consistent.

## 15. Notes

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| Purpose                | Command                                                             |
|------------------------|---------------------------------------------------------------------|
| Check Git version      | <code>git --version</code>                                          |
| Set user name          | <code>git config --global user.name "Your Name"</code>              |
| Set email              | <code>git config --global user.email "youremail@example.com"</code> |
| Default branch         | <code>git config --global init.defaultBranch main</code>            |
| Check configuration    | <code>git config --list</code>                                      |
| Set credential manager | <code>git config --global credential.helper manager</code>          |
| Set PATH (if needed)   | Add <code>C:\Program Files\Git\cmd</code> to environment variables  |